higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

18 April 2024

This marking guideline consists of 7 pages.

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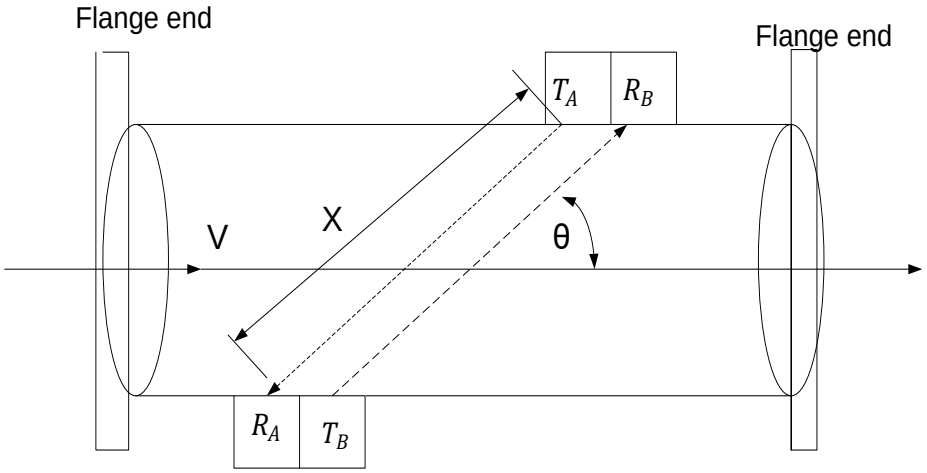
SECTION A

QUESTION 1

1.1
- Square root compensation
- Square root extraction (2)

1.2
- Mechanical
- Electrical
- Pneumatic (3)

1.3



(1 mark for each transmitter/receiver pair. 2 marks for indicating angle and distance between the transducer pair.1 mark for the sketch.) (5)

1.4 Advantages
- Low manufacturing costs
- Can be machined very accurately
- Easily installed
- Maintenance is free (Any 3 × 1)

Disadvantages
- High pressure loss
- It cannot be used with liquids containing heavy material, which can form a layer of dirt.
- In case of an area ratio bigger than 0.55, the measurement will be inaccurate. (Any 2 × 1) (5)

[15]

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QUESTION 2

2.1

- It was proven by Bernoulli that flow varies proportional to the square root of the differential pressure across an orifice plate.
- This differential pressure is measured by a differential pressure sensitive device, therefore measuring the differential pressure and not the flow.
- If the flow is to be determined, the measurement should be displayed as the square root of the differential pressure.
- The problem of compensation for this square root is known as the square root problem.

OR

The primary measuring device is used to create a differential pressure in a flowing fluid.✓ A secondary device then measures this difference in pressure to determine the volume flow rate of the fluid.✓ However, it was discovered by Bernoulli that the flow rate is not directly proportional to the differential pressure✓ but it is proportional to the square root of differential pressure.✓ Hence the equation: $Q = k\sqrt{\Delta P}$,✓ with k being the proportionality constant. (4)

2.2

- This apparatus consists of a long, graduated tube having a uniform taper.
- The metering tube is mounted vertically with the small end at the bottom.
- A float in the form of a plumb bob moves freely up and down along the axis of the tube.
- As the flow rate through the instrument increases, the float rises in the tube.
- As the float rises and the area around the float increases, it keeps the differential pressure across the float constant.
- The position of the float is then used as an indication of the flow through the rotameter and is read of on the graduated portion of the tube.

OR

Consider a vertical tube of a conical shape, the area gradually expanding from bottom to the top.√ In the tube is a fluid flowing in an upward direction and in it is placed a disc, free to move,√ so that it acts as a sort of float in the fluid.√ An orifice is set up between the disc and the inlet of the tube,√ thus a differential pressure is generated across it.√ Certain forces act on the disc and these forces are at equilibrium.√ A change in flow will affect the pressure drop, changing the relation between the inlet and outlet pressure,√ upsetting the equilibrium of the forces acting on the disc.√ The disc will then move up or down the tube,√ adjusting the area of the orifice due to the conical shape of the tube,√ until the pressure drop is at the original value.√ The position of the float in the tube is then a measure of the flow rate.√ (6)

2.3 The reading is taken at the centre of the float;✓ it is recommended that the float be at eye level to minimise reading errors.✓ (2)

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2.4
- An integral orifice cell is a miniature flow mounted to a DP cell.✓✓
- They can be used for laboratory scale processes to measure additives to major flow streams; or in general it is used for the measurement of small fluid flows.✓ (3)

2.5
2.5.1 C
2.5.2 E
2.5.3 A
2.5.4 F
2.5.5 B
2.5.6 D

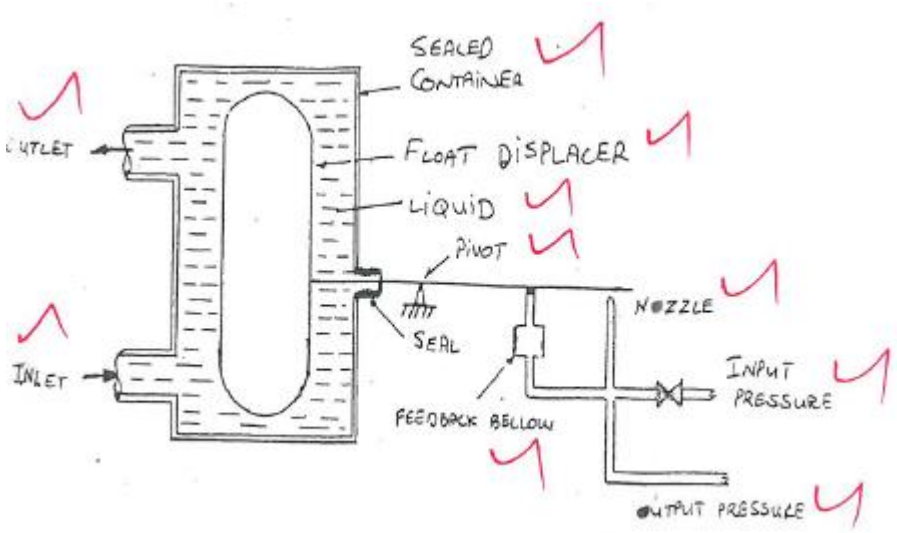
(6 × 1) (6)
[21]

TOTAL SECTION A: 36

SECTION B

QUESTION 3: DENSITY, HUMIDITY & VISCOSITY MEASUREMENT

3.1



(5)

3.2 If the temperature of the flowing liquid changes, a thermostatic heater may be used to hold it constant. (2)

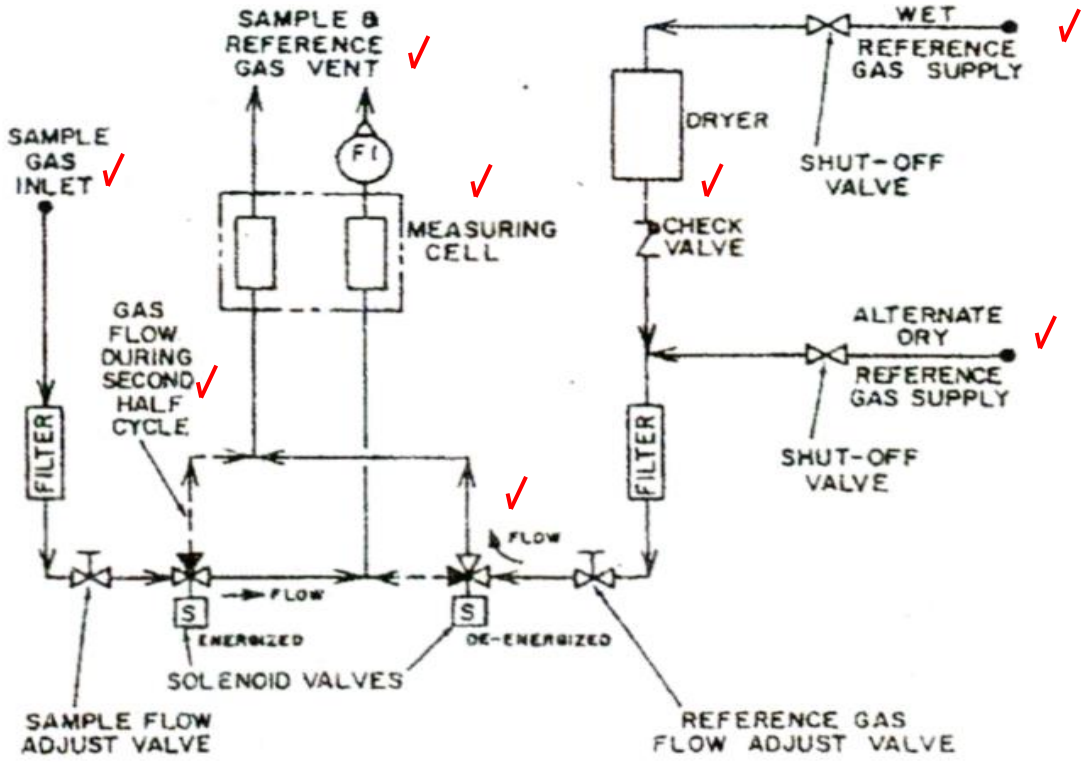
3.3 3.3.1 Capillary tube viscometer (1)

3.3.2 A – Filling etched line
B – Lower etched index line
C – Efflux vessel
D – Capillary tube
E – Receiving vessel
F – Upper etched index line (6)

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3.4
- Single float
- Two float
- Concentric float (3)

3.5



(8)
[25]

QUESTION 4: PH AND CONDUCTIVITY MEASUREMENT

4.1
- pH-sensitive glass membrane electrode
- A reference electrode
- A high input impedance (3)

4.2
- The ideal housing will have resistance to corrosion.✓
- Strength against thermal✓ and mechanical shock✓ (3)

4.3
- The total immersion type
- The continuous flow type (2)

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4.4
- Treatment with strong oxidising agents such as hot chromic acid or nitric acid solution is effective for preparing platinum, gold and tungsten electrodes.
- Thorough rinsing is required following treatment.
- Cleaning with a fine abrasive is sometimes required for metal electrodes.
- The electrode activity is generally more stable when electrodes are stored under water. (4)

4.5 4.5.1 By adding a regulated power supply to drive a transformer

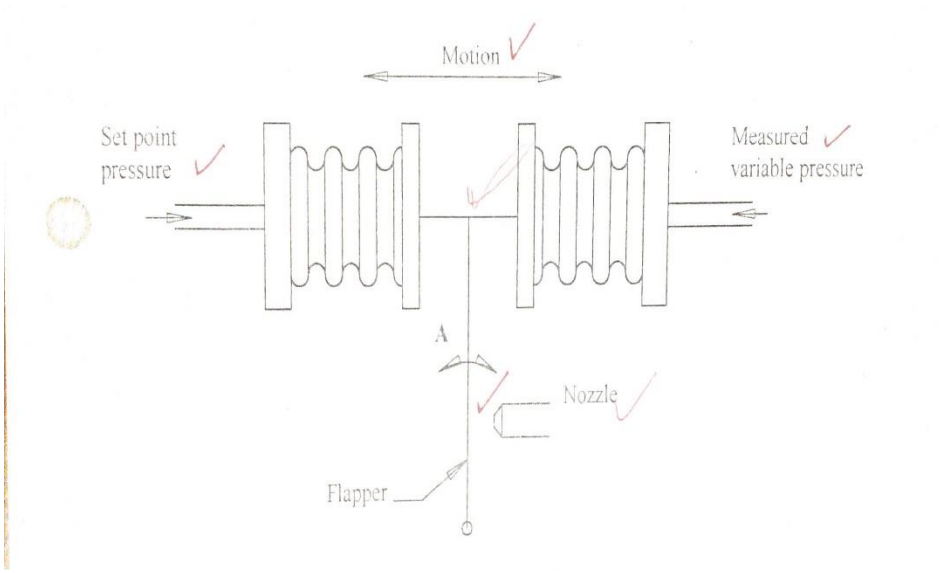
4.5.2 A stage of amplification is added✓✓ to increase sensitivity and reduce non-linearity.

(2 × 2) (4)
[16]

QUESTION 5: AUTOMATIC CONTROL

5.1
- Distance velocity lags
- Transfer lags
- Process lags
- Measurement lags (4)

5.2 5.2.1



(6)

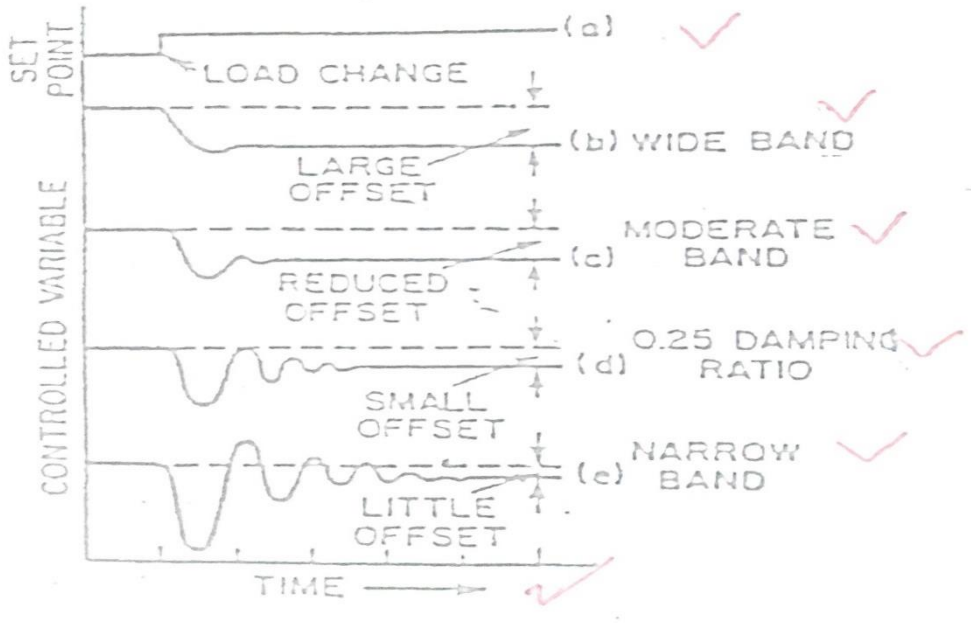
5.2.2 The set point pressure must equal the measured variable pressure to balance the system. (2)

5.2.3 When the measured variable pressure changes, it causes the bellow to expand✓ or contract. This will change the position of the lever.✓ A flapper nozzle system is connected to the lever to rectify the deviation.✓ (3)

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5.3 Integral control (2)

5.4



One can see the oscillation's amplitude starts to increase with narrow band. (6)
[23]

TOTAL SECTION B: 64
GRAND TOTAL: 100

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